Semiconductor & Quantum Patents in CN and TW (2024–2026)

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1. Executive summary

80,062 CN and 19,484 TW invention publications in semiconductors and quantum technologies were first published between 1 Nov 2024 and the data cut-off (CN: 8 Sep 2026; TW: Jun 2026). China is a mostly domestic, broad-based and academically heavy filing system. Taiwan is a small, foreign-dominated market led by TSMC and the global equipment makers.

Data findings

- **The corpus is far larger than the EP-style scope suggests.** Adding IPC to CPC and including the new 2026 subclasses H10P (processes) and H10W (packaging/interconnect) raises the CN corpus from 52,987 to 80,062 (+51%) and TW from 13,650 to 19,484 (+43%). With the legacy scope, CN would show a false 46% collapse in Q1 2026.
- **Largest technology fields (CN):** wafer handling & process apparatus 晶圆传输与工艺装置 (18,596), package substrates & leadframes 封装基板与引线框架 (8,611), deposition & epitaxy 薄膜沉积与外延 (7,607), bumps/wire bonds 凸块/引线键合 (7,193), encapsulation 塑封 (7,094). TW has the same top tier, led by wafer handling (5,659) and deposition (3,829).
- **Fastest-growing (CN, like-for-like Y1→Y2, full text):** 3D stacking 三维堆叠 +98%, chiplets/2.5D/interposer 芯粒/中介层 +94%, hybrid bonding 混合键合 +82%, glass substrates/TGV 玻璃基板/玻璃通孔 +82%, wafer bonding & layer transfer +48%, fan-out packaging 扇出型封装 +38%. Advanced packaging is the clear growth front.
- **Top players.** CN: Samsung (3,410), Hua Hong (1,454), TSMC (1,201), Chinese Academy of Sciences (1,182), BOE (1,135), Applied Materials (951), Tokyo Electron (909), YMTC (893), Huawei (888). TW: TSMC (2,083), Applied Materials (1,252), Tokyo Electron (1,042), Samsung (1,026), Nanya (550).
- **Origin.** 69% of CN filings are domestic (55,329). Japan 8,046, US 5,331, Korea 5,246, Taiwan 3,392 and Europe 2,635 follow. In TW, only 30% are Taiwanese (5,776): Japan 6,115, US 3,367, Korea 1,771, mainland China 1,090, Europe 1,005.
- **Quantum is a CN story.** 4,852 CN quantum inventions (6.1% of corpus, +21% Y1→Y2, 91% domestic, 39% academic) against 118 in TW. Origin Quantum 本源量子 (441) leads, then CAS, USTC, the Yangtze River Delta Quantum Innovation Center and Hefei

National Laboratory. The Anhui/Hefei cluster dominates. Algorithms & software 量子算法与软件 is the biggest sub-field (3,250).

- **Academic weight.** Universities or research institutes appear on 13.6% of CN filings, against 2.1% in TW.
- **Cross-strait.** TSMC, Nanya, UMC, Innolux and ASE file actively in CN (3,392 CN publications with TW applicants). Mainland equipment makers (Naura, ACM Research, AMEC) and JCET are the main CN filers in TW (1,090). 55% of TW publications belong to families that also have a CN member.

Main caveats: CN applicant nationality is not in the data and is derived from names and priority claims. TW 2026 volumes are not robust (see section 2).

2. Data, scope and corpus

The corpus covers invention patents (kind A/B) whose first publication falls between 2024-11-01 and the last available date. Each application is counted once.

Data freshness (Google Patents Public Datasets, queried 3 Nov 2026): CN latest publication 2026-09-08, so September is partial and August is the last full month. TW latest overall is 2026-09-01, but TW A/B publications stop after June 2026; July–October are missing, not zero.

Measure	CN	TW
Invention publications (applications)	80,062	19,484 (A 17,231 · B 2,253)
Distinct families	79,118	18,824
PCT national phase	9,524 (11.9%)	0 (Taiwan is not a PCT state)
PCT-excluded variant (EP-comparable): applications / families	70,538 / 69,783	19,484 / 18,824
Utility models, same scope (separate line, not in totals)	18,288 (CN U, 实用新型)	839 (TW U, 新型)
Selected only via IPC (no CPC match)	28,982 (36.2%)	3,967 (20.4%)
Would be in the EP-study (legacy) scope	52,987	13,650
Abstract retrieved in the second run (4 Nov 2026)	23,072 (28.8%)	3,866 (19.8%)
Family also filed at EP / US / JP / KR	19,783 (24.7%)	13,296 (68.2%)

Measure	CN	TW
Family also filed at EP	5,002 (6.2%)	2,284 (11.7%)

Filing-date view. The median filing month of documents first published in Nov 2024 is Jul 2024 for CN and Jan 2024 for TW. By Aug 2026 it is Apr 2026 for CN. The window therefore mostly covers filings from mid-2024 to early 2026; CN publishes faster than TW.

Kind codes. CN first publications are almost all A (one B). CN U utility models are reported separately. TW has A, B (grants without prior A publication), U/M utility models and S designs (excluded).

Why the corpus differs from the EP-style scope. Two data facts force deviations, which are applied to both offices:

1. 31.6% of all CN invention publications in the window carry no CPC at all, and CPC on CN documents is a partial mapping. IPC is 100% populated, so selection uses **CPC ∪ IPC**.
2. From 2026 the data carries the new subclasses **H10P** (generic processes, successor of H01L 21/) and **H10W** (packages, interconnects, isolation, successor of H01L 23/25). CN documents carry no H01L at all from January 2026. Both subclasses were added to the core scope. Note: the EP prompt's regex misses them too and will under-count 2026.

3. Technology rankings

Process equipment and packaging dominate both offices by volume. Wafer handling & process apparatus alone is 23% of CN and 29% of TW filings. Counting is **full**: a document counts in every category it matches, so shares add up to more than 100%. Change = like-for-like windows (CN Nov–Aug vs Nov–Aug; TW Nov–Jun vs Nov–Jun), computed on full text (codes, title and abstract) for both periods. Baselines: CN total +2.5% (35,539 → 36,428), TW total −11% (8,104 → 7,207).

CN (CNIPA) — top 22 of 50 core categories

Technology	Chinese term	Code basis	Apps	Families	Share	Change	Top 5 applicants (apps)
Wafer handling & process apparatus	晶圆传输与工艺装置	H01L21/67–68, H10P72	18,596	18,362	23.2%	−5%	Tokyo Electron 704; Naura 608; Samsung 593; Applied Materials 541; SCREEN 363
Package substrates & leadframes	封装基板与引线框架	H01L23/49, H10W70 + text	8,611	8,501	10.8%	+4%	Samsung 391; JCET 247; TSMC 191; Huawei 180; Infineon 167
Deposition & epitaxy	薄膜沉积与外延	H01L21/02, H10P14 + text	7,607	7,530	9.5%	−25%	Applied Materials 389; Tokyo Electron 271; ASM 232; Hua Hong 214; Lam 176
Bumps, wire bonds & package interconnects	凸块/引线键合/封装互连	H10W72 + text	7,193	7,064	9.0%	−10%	Samsung 380; JCET 214; TSMC 195; Infineon 144; Huawei 123
Encapsulation & housings	塑封与封装外壳	H10W74/76 + text	7,094	6,984	8.9%	−7%	Samsung 270; JCET 243; Infineon 163; Huawei 152; TSMC 139
CMP & cleaning	化学机械抛光与清洗	H01L21/304–321, H10P52/70 + text	6,494	6,385	8.1%	−10%	Tokyo Electron 259; Hua Hong 198; SCREEN 193; Applied Materials 168; Samsung 126
Thermal management of chips/packages	芯片与封装散热	H01L23/34–473, H10W40 + text	6,064	5,975	7.6%	+5%	Huawei 149; Samsung 124; JCET 106; TSMC 97; Infineon 79

Technology	Chinese term	Code basis	Apps	Families	Share	Change	Top 5 applicants (apps)
Etching	刻蚀	H01L21/3065–311, H10P50 + text	5,912	5,849	7.4%	−10%	Tokyo Electron 344; Hua Hong 307; Samsung 176; Applied Materials 169; TSMC 129
On-chip interconnect (BEOL)	片上互连(后道)	H01L21/768, H01L23/52, H10W20	5,352	5,258	6.7%	−14%	Samsung 590; TSMC 340; SK hynix 173; Nanya 161; YMTC 135
In-line test & metrology	在线检测与量测	H01L21/66, H01L22, H10P74	5,092	5,052	6.4%	+31%	Hua Hong 146; Applied Materials 127; Samsung 111; Eswin 106; Tokyo Electron 87
IC design / EDA	集成电路设计/EDA	G06F30/39	4,425	4,410	5.5%	+15%	Inspur 118; CAS 94; State Grid 64; UESTC 62; Huawei 58
Thin-film transistor (TFT)	薄膜晶体管	H01L29/786, H10D86 + text	3,408	3,322	4.3%	−19%	BOE 612; Samsung 479; LG Display 356; Tianma 214; TCL CSOT 195
Doping & annealing	掺杂与退火	H01L21/22–324, H10P30/32 + text	3,378	3,356	4.2%	−9%	Hua Hong 144; CAS 94; Applied Materials 92; Shanghai Jita 62; TSMC 61
Quantum algorithms & software	量子算法与软件	G06N10/20–80 + text	3,250	3,237	4.1%	+27%	Origin Quantum 329; State Grid 73; China Mobile 72; CAS 71; USTC 70

Technology	Chinese term	Code basis	Apps	Families	Share	Change	Top 5 applicants (apps)
DRAM	动态随机存取存储器	H10B12, H01L27/108 + text	2,978	2,906	3.7%	+7%	Samsung 548; YMTC 373; CXMT 354; Beijing Superstring Academy 315; Nanya 282
Lithography & patterning (process)	光刻与图形化	H01L21/027–033, H10P76 + text	2,805	2,778	3.5%	−6%	Hua Hong 165; Samsung 132; Tokyo Electron 106; Applied Materials 80; CAS 69
NAND / 3D NAND flash	NAND/三维闪存	H10B41/43, H01L27/115 + text	2,553	2,491	3.2%	+5%	YMTC 406; SK hynix 282; Samsung 262; Hua Hong 234; Kioxia 210
SiC devices	碳化硅器件	text in scope	2,067	2,052	2.6%	+5%	Xidian Univ. 73; Global Power Tech 66; Gree 49; CAS 41; Hangzhou Puxi 35
Display backplanes / array substrates	显示面板阵列基板	text in scope	1,911	1,839	2.4%	−39%	BOE 664; Visionox 246; Tianma 243; TCL CSOT 207; Samsung 112
Through-silicon via (TSV)	硅通孔	H10W20/2 + text	1,865	1,846	2.3%	−22%	Samsung 269; TSMC 108; Intel 69; Huawei 49; SK hynix 38
GaN devices / HEMT	氮化镓器件/HEMT	H01L29/7786 + text	1,628	1,612	2.0%	+7%	Xidian Univ. 207; CAS 70; UESTC 40; SITRI 35

Technology	Chinese term	Code basis	Apps	Families	Share	Change	Top 5 applicants (apps)
Si power MOSFET (trench/SJ/LDMOS)	硅功率 MOSFET(沟槽/超结)	text in scope	1,547	1,542	1.9%	−2%	Hua Hong 158; China Resources Micro 63; UESTC 52; Global Power Tech 38

TW (TIPO) — top 17

Technology	Chinese term	Apps	Families	Share	Change	Top 5 applicants (apps)
Wafer handling & process apparatus	晶圓傳送與製程裝置	5,659	5,510	29.0%	−19%	Tokyo Electron 644; Applied Materials 629; SCREEN 285; Lam 245; TSMC 225
Deposition & epitaxy	薄膜沉積與磊晶	3,829	3,743	19.7%	−32%	Applied Materials 468; TSMC 372; Tokyo Electron 345; Shin-Etsu 180; Lam 177
On-chip interconnect (BEOL)	晶片內連線(後段)	2,615	2,482	13.4%	−21%	TSMC 643; Samsung 323; Nanya 181; Applied Materials 152; UMC 84
CMP & cleaning	化學機械研磨與清洗	2,398	2,350	12.3%	−19%	Tokyo Electron 339; Applied Materials 212; SCREEN 181; TSMC 143; Ebara 75
Bumps, wire bonds & package interconnects	凸塊/打線接合/封裝互連	2,373	2,270	12.2%	−13%	TSMC 427; JCET 105; Samsung 104; Amkor 104; Qualcomm 83
Etching	蝕刻	2,297	2,249	11.8%	−9%	Tokyo Electron 447; TSMC 271; Applied Materials 228; Lam 117; Nanya 109
Package substrates & leadframes	封裝基板與導線架	2,185	2,111	11.2%	−17%	TSMC 382; Amkor 105; Samsung 104; JCET 101; ASE 94
Encapsulation & housings	封膠與封裝外殼	1,935	1,869	9.9%	−12%	TSMC 320; Amkor 109; ASE 106; JCET 105; Samsung 68
Lithography & patterning (process)	微影與圖案化	1,786	1,738	9.2%	−26%	Tokyo Electron 169; Shin-Etsu 164; TSMC 146; Samsung 116; Applied Materials 92

Technology	Chinese term	Apps	Families	Share	Change	Top 5 applicants (apps)
In-line test & metrology	線上檢測與量測	1,177	1,140	6.0%	−11%	Applied Materials 136; Tokyo Electron 92; TSMC 78; KLA 62; Samsung 46
Thermal management	晶片與封裝散熱	1,143	1,113	5.9%	−9%	TSMC 186; ASE 52; Adeia 38; JCET 25; onsemi 24
DRAM	動態隨機存取記憶體	998	904	5.1%	+25%	Nanya 316; Samsung 237; Semiconductor Energy Lab 58; Winbond 58; Applied Materials 49
Doping & annealing	摻雜與退火	811	788	4.2%	−17%	TSMC 138; Applied Materials 125; Nanya 30; UMC 23; Tokyo Electron 22
Through-silicon via (TSV)	矽穿孔	772	742	4.0%	−19%	TSMC 215; Samsung 147; Nanya 36; Amkor 27; UMC 26
NAND / 3D NAND flash	NAND/3D快閃記憶體	720	663	3.7%	+5%	Kioxia 193; Samsung 59; Macronix 56; Winbond 46; Semiconductor Energy Lab 41
Hybrid bonding	混合鍵合	573	549	2.9%	−19%	TSMC 160; Adeia 57; Samsung 39; Applied Materials 28; Kioxia 27
Thin-film transistor (TFT)	薄膜電晶體	542	506	2.8%	−39%	Semiconductor Energy Lab 132; LG Display 49; TSMC 46; Samsung 45; AUO 35

Not categorised beyond domain level: 8,388 CN (10.5%) and 1,794 TW (9.2%) documents. Code basis is the same for TW as for CN. The full 65-category tables with basis counts (code only / code + text / text only) are in the BigQuery table `cntw_study.application_list`.

4. Per-domain breakdown

Domain 1 (transistors, devices, memory and processes) is the largest in both offices: CN 56,974 · TW 16,163. Domain 2 (chips, integration & design) is CN 30,220 · TW 7,663. Quantum is CN 4,852 · TW 118. Domains overlap: one document can sit in several.

Domain 1 — devices, materials and memory (beyond the top tables)

Sub-technology	Chinese term (CN / TW)	CN	CN change	TW	Leading applicants (CN)
Dicing & singulation	切割与划片 / 切割與晶粒分離	1,728	+4%	478	Resonac 50; Samsung 37; Hua Hong 24
Device isolation & buried regions	器件隔离与埋层 / 元件隔離與埋層	1,426	−8%	449	Hua Hong 165; Nexchip 67; TSMC 58
IGBT	绝缘栅双极型晶体管 / 絕緣閘雙極電晶體	1,289	−16%	96	Mitsubishi Electric 79; Infineon 72; Toshiba 72; Fuji Electric 51
RRAM / memristor	阻变存储器/忆阻器 / 電阻式記憶體	1,110	+6%	262	SK hynix 133; Xincun Tech 99; Samsung 91; UMC 45
Oxide semiconductor (IGZO)	氧化物半导体 / 氧化物半導體	891	−5%	273	Samsung 70; Semiconductor Energy Lab 59; Japan Display 51
Wafer bonding & layer transfer	晶圆键合与层转移 / 晶圓接合與層轉移	851	+48%	246	CAS 34; Tokyo Electron 28; Piotech 23; Soitec 21
Ferroelectric (FeFET/FeRAM)	铁电器件 / 鐵電元件	828	+14%	146	CAS 67; Samsung 61; Huawei 46; YMTC 46
MRAM / spintronics	磁性随机存储器 / 磁阻式記憶體	699	−5%	259	Samsung 93; SK hynix 65; Hikstor 64; UMC 45
Image sensors	图像传感器 / 影像感測器	608	−48%	217	Sony 103; Hua Hong 28; Samsung 26
2D-material devices	二维材料器件 / 二維材料元件	548	+13%	84	CAS 37; Samsung 19; HUST 17; Fudan 15
GAA / nanosheet	全环绕栅极/纳米片晶体管 / 環繞式閘極電晶體	396	+8%	129	CAS 53; Samsung 48; Intel 24
CFET / stacked transistors	互补场效应晶体管 / 互補式場效電晶體	392	+8%	113	Peking Univ. 46; TSMC 42; CAS 31; Superstring Academy 28
Ga2O3 devices	氧化镓器件 / 氧化鎵元件	341	+20%	22	Xidian Univ. 59; CAS 19; Jiufengshan Lab 16
Micro-/Mini-LED	微型发光二极管 / 微型發光二極體	243	−18%	60	ams-OSRAM 21; LG Display 15; JBD 8
III-V (GaAs/InP)	砷化镓/磷化铟器件	237	+30%	48	CAS 12; Guangdong Xiandao 11; Innoscience 9
FinFET	鳍式场效应晶体管 / 鰭式場效電晶體	156	0%	103	Samsung 21; Xidian 11; CAS 9; SMIC 8; TSMC 8
Phase-change memory	相变存储器 / 相變記憶體	100	−26%	10	Xincun Tech 32; CAS 9
Compute-in-memory	存算一体 / 記憶體內運算	82	+40%	3	HUST 6; CAS 6; Peking Univ. 5

Material platforms in CN, by count: SiC 2,067 > GaN 1,628 > oxide 891 > 2D 548 > Ga2O3 341 > III-V 237. Wide-bandgap power work is driven by universities (Xidian, UESTC, CAS) and domestic power firms. In TW, SiC (149) and GaN (113) are small and mostly materials suppliers.

Domain 2 — advanced packaging and integration

Sub-technology	Chinese term (CN / TW)	CN	CN change	TW	TW change	Leading applicants
Chiplets / 2.5D / interposer	芯粒/中介层 / 小晶片/中介層	1,343	+94%	386	+10%	CN: TSMC 54, JCET 53, Samsung 53, Intel 39 · TW: TSMC 100, JCET 30
3D stacking / monolithic 3D	三维堆叠 / 三維堆疊	1,284	+98%	279	+45%	CN: Samsung 121, YMTC 40, SK hynix 37, Intel 37 · TW: TSMC 59, Samsung 29
Hybrid bonding	混合键合 / 混合鍵合	1,242	+82%	573	−19%	CN: Samsung 99, YMTC 69, SK hynix 64, TSMC 61 · TW: TSMC 160, Adeia 57
Fan-out / wafer- & panel-level	扇出型/晶圆级封装 / 扇出型/晶圓級封裝	1,174	+38%	270	−27%	CN: Huatian 51, Samsung 37, Huawei 32, JCET 30 · TW: TSMC 55
Power modules	功率模块 / 功率模組	1,137	+7%	42	−20%	CN: Huawei 66, Changfei 30, Hyundai 25, Kia 25
Glass substrates / TGV	玻璃基板/玻璃通孔	584	+82%	97	+32%	CN: Absolics (Ebersolic) 31, Intel 31, Visionox 30, Chengdu Yicheng 29 · TW: Absolics 29, NEG 9
Backside power delivery	背面供电 / 背面供電	110	−25%	69	−25%	CN: IBM, Samsung, SJSemi · TW: Samsung, TSMC

Domain 3 — quantum (corrected definition, see section 11)

Sub-technology	Chinese term (CN / TW)	CN	CN change	TW	Leading applicants (CN)
Algorithms & software	量子算法与软件 / 量子演算法與軟體	3,250	+27%	60	Origin Quantum 329; State Grid 73; China Mobile 72; CAS 71; USTC 70
Superconducting qubits & devices	超导量子比特 / 超導量子位元	608	+6%	31	Origin Quantum 67; CAS 43; YRD Quantum Center 36; BAQIS 34; Hefei Nat. Lab 31

Sub-technology	Chinese term (CN / TW)	CN	CN change	TW	Leading applicants (CN)
Cryogenics, control & readout	低温、测控与读取	509	+25%	13	Origin Quantum 93; USTC 37; Hefei Nat. Lab 33; YRD Center 33; SpinQ 29
Error correction / mitigation	量子纠错与误差缓解	467	+19%	18	Origin Quantum 59; Google 26; USTC 17
Coupling & quantum-chip architecture	量子比特耦合与量子芯片架构	403	+32%	6	Origin Quantum 130; YRD Center 36; SpinQ 22; Inspur 21
Photonic quantum computing	光量子计算	96	+340%	1	Turing Quantum (Shanghai/Wuxi) 38; Bose Quantum 17
Trapped-ion	离子阱	82	0%	1	USTC 16; Hefei Yaozheng Quantum 10; Huawei 9
Neutral atoms	中性原子	55	+74%	0	Zhongke Kuyuan 5; Huawei 5; BAQIS 4
Spin / semiconductor qubits	自旋/半导体量子比特	38	−13%	2	RWTH Aachen 10; FZ Jülich 8; Hefei Nat. Lab 5
Topological	拓扑量子	21	+86%	2	SJTU 2; Microsoft 2

TW quantum (118) is almost entirely foreign: IBM 10, Multiverse Computing (ES) 9, Photonic Inc. (CA) 8, Silicon Quantum Computing (AU) 5, ITRI 5.

Extensions (reported separately; full extension sets, not only the core overlap)

Extension	Chinese term	CN	CN change	TW	Leading applicants (CN)
QKD / quantum cryptography	量子密钥分发 / 量子金鑰分發	1,760	−34%	10	China Telecom Quantum 156; State Grid 82; Matrix Time 60; China Mobile 56; QuantumCTek 44
Quantum communication & networks	量子通信与网络	1,006	−16%	25	China Telecom 59; QuantumCTek 31; USTC 30; CAS 30
Quantum magnetometry (NV/atomic/SQUID)	量子磁测	689	+46%	6	Beihang Univ. 119; State Grid 31; Anhui Guosheng Quantum 23
Atomic clocks	原子钟	260	+46%	0	CAS 59; Beijing Inst. Radio Metrology 31; Peking Univ. 12
Lithography tools, masks & resists	光刻设备/掩模/光刻胶	9,690	+11%	3,634	ASML 497; CAS 358; Shin-Etsu 308; Samsung 251; Hua Hong 187

Extension	Chinese term	CN	CN change	TW	Leading applicants (CN)
CVD / ALD	化学气相沉积/原子层沉积	8,283	+4%	2,206	Applied Materials 345; Piotech 319; ASM 296; CAS 246; Lam 213
Plasma processing equipment	等离子体工艺设备	3,318	+6%	1,837	Applied Materials 346; Tokyo Electron 343; Naura 196; Lam 194
Wafer handling apparatus	晶圆处理装置	3,270	+51%	1,360	Naura 149; Samsung 116; Applied Materials 82; Tokyo Electron 80; Piotech 65

Extension totals: quantum comm. & sensing CN 4,362 / TW 44; equipment CN 35,011 / TW 10,679. G01N24 (magnetic resonance, CN 1,282) is dominated by oil-industry NMR (PetroChina 119) and is not quantum-specific. C23C16 also includes non-semiconductor coatings.

5. Top applicants

Concentration is low in CN and high in TW. The top 5 groups hold 10.5% of CN filings (8,382 of 80,062) but 30.6% of TW filings (5,953 of 19,484). Applicants are grouped under parents (rules in section 11). Change = like-for-like windows; compare against the office baselines (CN +2.5%, TW −11%).

CN — top 20

#	Applicant (parent)	Type	Country (CN region)	Apps	Y1 → Y2	Technology profile (top 3)
1	Samsung (incl. Display, SDI, SEMES)	company	KR	3,410	1,595 → 1,517 (−5%)	wafer handling; BEOL; DRAM
2	Hua Hong Group (incl. Huali) 华虹	company	CN (Shanghai)	1,454	563 → 733 (+30%)	etching; NAND; deposition
3	TSMC 台积电	company	TW	1,201	544 → 538 (−1%)	BEOL; bumps; package substrates
4	Chinese Academy of Sciences 中国科学院 (all institutes)	institute	CN (Beijing)	1,182	554 → 486 (−12%)	deposition; EDA; etching
5	BOE 京东方	company	CN (Beijing)	1,135	672 → 375 (−44%)	display backplanes; TFT
6	Applied Materials	company	US	951	451 → 413 (−8%)	wafer handling; deposition; etching

#	Applicant (parent)	Type	Country (CN region)	Apps	Y1 → Y2	Technology profile (top 3)
7	Tokyo Electron	company	JP	909	385 → 414 (+8%)	wafer handling; etching; deposition
8	YMTC 长江存储	company	CN (Hubei)	893	329 → 423 (+29%)	NAND; DRAM; BEOL
9	Huawei 华为 (incl. HiSilicon)	company	CN (Guangdong)	888	392 → 381 (−3%)	package substrates; thermal; encapsulation
10	SMIC 中芯国际	company	CN (Shanghai)	687	326 → 272 (−17%)	BEOL; metrology; NAND
11	Naura 北方华创	company	CN (Beijing)	648	266 → 283 (+6%)	wafer handling; CMP/cleaning; etching
12	CXMT 长鑫	company	CN (Anhui)	638	423 → 153 (−64%)	DRAM; BEOL; bumps
13	SK hynix	company	KR	603	267 → 283 (+6%)	NAND; BEOL; RRAM
14	Nexchip 晶合集成	company	CN (Anhui)	557	181 → 323 (+78%)	etching; isolation; metrology
15	Beijing Superstring Academy of Memory Tech. 超弦存储器研究院	institute	CN (Beijing)	546	225 → 278 (+24%)	DRAM; NAND; ferroelectric
16	Xidian University 西安电子科技大学	university	CN (Shaanxi)	538	281 → 219 (−22%)	GaN/HEMT; SiC; Ga2O3
17	LG Display	company	KR	531	243 → 275 (+13%)	TFT; display backplanes
18	Nanya Technology 南亚科技	company	TW	499	212 → 204 (−4%)	DRAM; BEOL; etching
19	Visionox 维信诺	company	CN (Jiangsu)	471	225 → 220 (−2%)	display backplanes; TFT; substrates
20	Origin Quantum 本源量子	company	CN (Anhui)	447	175 → 222 (+27%)	quantum algorithms; quantum chips; control & readout

Next five: Infineon (DE) 436, UESTC 423, SCREEN (JP) 382, JCET 379 (+24%), Shanghai Jita 376 (−70%).

TW — top 20

#	Applicant (parent)	Type	Country	Apps	Y1 → Y2	Technology profile (top 3)
1	TSMC 台積電	company	TW	2,083	937 → 811 (−13%)	BEOL; bumps; package substrates

#	Applicant (parent)	Type	Country	Apps	Y1 → Y2	Technology profile (top 3)
2	Applied Materials	company	US	1,252	528 → 469 (−11%)	wafer handling; deposition; etching
3	Tokyo Electron	company	JP	1,042	392 → 439 (+12%)	wafer handling; etching; deposition
4	Samsung (incl. Display, SDI, SEMES)	company	KR	1,026	419 → 390 (−7%)	BEOL; DRAM; TSV
5	Nanya Technology 南亞科技	company	TW	550	259 → 164 (−37%)	DRAM; BEOL; etching
6	SCREEN	company	JP	374	146 → 141 (−3%)	wafer handling; CMP/cleaning; etching
7	UMC 聯電	company	TW	359	134 → 131 (−2%)	BEOL; RRAM; MRAM
8	Lam Research	company	US	356	132 → 131 (−1%)	wafer handling; deposition; etching
9	Kioxia	company	JP	320	85 → 169 (+99%)	NAND; BEOL; bumps
10	Shin-Etsu	company	JP	299	130 → 99 (−24%)	deposition; lithography; etching
11	ASM International	company	NL	235	85 → 97 (+14%)	wafer handling; deposition
12	Semiconductor Energy Laboratory	company	JP	205	74 → 90 (+22%)	TFT; oxide semiconductor; DRAM
13	Resonac	company	JP	195	89 → 59 (−34%)	encapsulation; bumps; CMP
14	ASE Group (incl. SPIL) 日月光/矽品	company	TW	184	77 → 79 (+3%)	encapsulation; substrates; bumps
15	Naura 北方華創	company	CN	179	57 → 73 (+28%)	wafer handling; etching; deposition
16	Fujifilm	company	JP	170	76 → 46 (−39%)	deposition; lithography
17	Innolux 群創	company	TW	166	88 → 39 (−56%)	BEOL; substrates; encapsulation
18	Qualcomm	company	US	160	66 → 75 (+14%)	substrates; bumps; BEOL
19	Powerchip (PSMC) 力積電	company	TW	155	65 → 55 (−15%)	lithography; deposition; BEOL
20	Kokusai Electric	company	JP	154	61 → 51 (−16%)	deposition; wafer handling

Applicant types (share of filings with at least one applicant of that type):

Type	CN	TW
Company	69,856 (87.3%)	19,129 (98.2%)
University	7,529 (9.4%)	289 (1.5%)
Research institute	3,738 (4.7%)	127 (0.7%)
University or institute	10,861 (13.6%)	407 (2.1%)
Individual	393 (0.5%)	113 (0.6%)
No applicant field	0	0

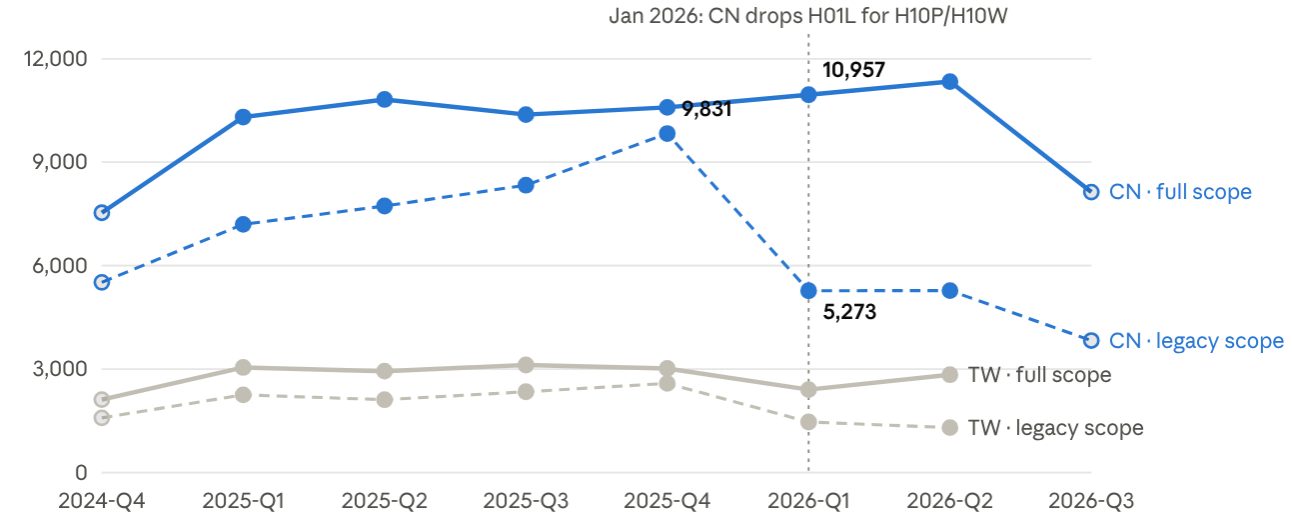
In CN quantum the academic share is 38.9% (1,887 of 4,852).

6. Quarterly time series

CN volume is flat at about 10.3–11.3k inventions per quarter through Q2 2026. The extended scope removes a classification artefact that would otherwise read as a collapse.

Adding H10P/H10W keeps CN volume steady; the legacy scope falsely halves 2026

Invention publications per quarter of first publication; solid = full scope, dashed = EP-style legacy scope



Light points = incomplete quarter (Nov–Dec 2024; CN Jul–8 Sep 2026). TW has no data after Jun 2026.

BigQuery, patents-public-data · CN/TW A+B first publications Nov 2024–Sep 2026

The publication-lag boundary is the last complete month: Aug 2026 for CN and Jun 2026 for TW. Quarters after that are missing, not zero. TW's 2026 dip (2,406 in Q1) coincides with the same H01L → H10P/H10W transition, while total TW publications of all technologies stayed flat (≈3,100–3,600 a month). Treat the TW 2026 decline as **not robust**.

Top 10 technologies by quarter, CN (incomplete quarters in brackets):

Technology	[24-Q4]	25-Q1	25-Q2	25-Q3	25-Q4	26-Q1	26-Q2	[26-Q3]
Wafer handling & apparatus	1,714	2,618	2,551	2,522	2,636	2,348	2,436	1,771
Package substrates & leadframes	826	1,108	1,143	1,115	934	1,316	1,287	882
Deposition & epitaxy	975	1,109	1,082	1,065	1,192	736	866	582
Bumps / wire bonds	840	1,024	927	1,007	645	1,047	990	713
Encapsulation	741	979	960	988	617	1,097	996	716
CMP & cleaning	642	891	862	973	971	753	840	562
Thermal management	512	789	810	798	778	823	857	697
Etching	655	762	850	781	823	669	812	560
BEOL interconnect	559	761	780	761	674	624	716	477
In-line test & metrology	398	639	552	617	706	778	835	567

Top 10 technologies by quarter, TW:

Technology	[24-Q4]	25-Q1	25-Q2	25-Q3	25-Q4	26-Q1	26-Q2
Wafer handling & apparatus	623	912	901	915	913	642	753
Deposition & epitaxy	473	680	599	648	729	309	391
BEOL interconnect	334	410	441	378	326	296	430
CMP & cleaning	270	401	363	380	460	238	286
Bumps / wire bonds	278	373	376	348	303	295	400
Etching	242	376	345	331	371	274	358
Package substrates	233	359	360	346	269	252	366
Encapsulation	223	325	287	268	259	241	332
Lithography & patterning	224	311	250	317	350	127	207
In-line test & metrology	120	184	175	208	194	106	190

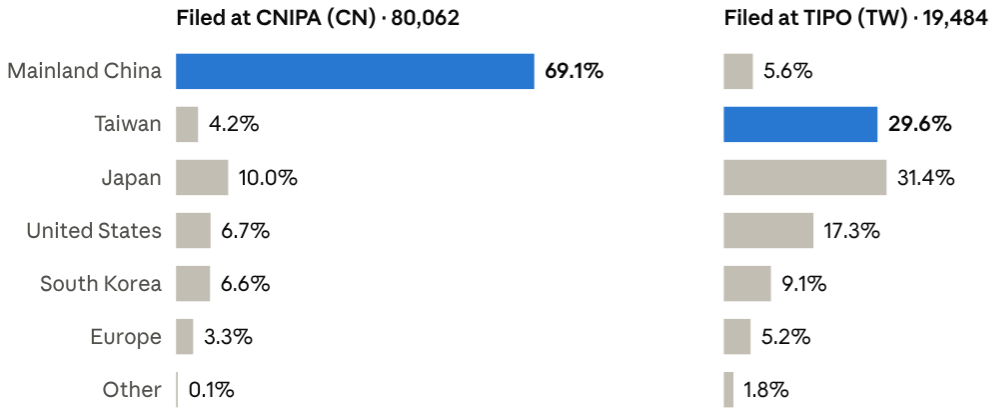
The Q4 2025 dips in CN bumps and encapsulation, and the 2026 drops in deposition, CMP and lithography, line up with months where CPC is missing or reclassified. Read them as classification effects until confirmed.

7. Origin of filings

CNIPA is a home market: 69% of filings are domestic. TIPO is a foreign-majority market: 70% of filings come from abroad, Japan first.

Mainland filings are 69% domestic; in Taiwan, Japanese applicants outnumber Taiwanese ones

Share of invention publications by applicant origin; highlighted = the office's home market



CN origin: names where known, else earliest priority office (inferred). TW origin: TIPO applicant country.

BigQuery, cntw_study.corpus · CN 80,062 / TW 19,484 invention publications

a) Priority office (earliest priority claim — data-derived for both offices)

Priority office	At CN	of which PCT route	At TW
Mainland China	55,696 (69.6%)	672	1,108 (5.7%)
United States	8,256 (10.3%)	2,985	7,441 (38.2%)
Japan	7,563 (9.4%)	4,167	5,809 (29.8%)
South Korea	5,330 (6.7%)	644	1,653 (8.5%)
Taiwan	1,009 (1.3%)	0	2,759 (14.2%)
EPO	825	456	263
Germany	669	225	133
France	254	135	78
United Kingdom	118	73	67

Many Taiwanese applicants (TSMC above all) claim **US first filings**, which is why only 14% of TW filings carry a TW priority although 30% have TW applicants.

b) Filing route

CN: 9,524 PCT national-phase entries (11.9%), mostly from Japanese (4,455) and US (2,797) applicants. Domestic CN applicants almost never use the PCT route to enter CN (695). TW: 0, since Taiwan is not a PCT contracting state and foreign filers enter via Paris Convention priority.

c) Europe

European applicants file 2,635 CN (3.3%) and 1,005 TW (5.2%) inventions. In CN the largest are Infineon 436, ASM International, Nexperia, STMicroelectronics and imec. In TW they are ASML, ASM International, Merck KGaA, Soitec and Zeiss. Europe's quantum filings in CN number 161 (e.g. RWTH Aachen and FZ Jülich on spin qubits).

d) Cross-strait flows

TW applicants filing in CN	CN apps	Y1 → Y2	CN applicants filing in TW	TW apps	Y1 → Y2
TSMC	1,201	544 → 538	Naura 北方华创	179	57 → 73
Nanya Technology	499	212 → 204	JCET 长电 (incl. STATS ChipPAC)	132	47 → 57
UMC	309	144 → 135	ACM Research 盛美 (Shanghai)	120	41 → 45
Innolux	246	144 → 83	Jade Bird Display 显耀	104	77 → 14
ASE Group	182	88 → 76	AMEC 中微	80	35 → 28
Winbond	144	39 → 86	AaltoSemi 芯爱	40	15 → 16
Macronix	86	51 → 20	YMTC 长江存储	36	16 → 14
Powerchip (PSMC)	82	30 → 40	CXMT 长鑫	22	6 → 15
Total TW-applicant filings in CN	**3,392**	1,581 → 1,434	**Total CN-applicant filings in TW**	**1,090**	486 → 368

Family links run deeper than direct filings. 55% of TW publications (10,795) have a CN family member, and 12% of CN publications (9,936) have a TW family member. Mainland firms entering Taiwan are overwhelmingly **equipment makers** (Naura, ACM, AMEC) and the packaging house JCET.

e) Inventor countries

CN: inventor country is empty for every record, so it cannot be compared. TW: populated for 100% of records, giving a data-derived view of where the research is done.

TW applicant origin	Apps	Inventors include TW	Inventors include mainland CN	Inventors include US	Multi-country inventor teams
Japan	6,115	53	119	233	366
Taiwan	5,776	5,637	83	354	580
United States	3,367	318	**771 (23%)**	2,814	2,022
South Korea	1,771	5	26	131	158

TW applicant origin	Apps	Inventors include TW	Inventors include mainland CN	Inventors include US	Multi-country inventor teams
Mainland China	1,090	133	866	112	257
Germany	287	24	26	69	127

US companies' Taiwanese filings often come from mainland-China R&D teams, and 318 come from teams in Taiwan. Taiwanese applicants' inventions are almost entirely Taiwan-made (98%).

f) CN regions — INFERRED from known headquarters of the top 50 domestic applicants

The dataset holds no applicant address. These 50 applicants cover 18,597 filings (34% of domestic CN filings).

Region	Filings (top-50 applicants)	Main applicants
Beijing	4,631	CAS, BOE, Naura, Superstring Academy, Peking Univ., Tsinghua, State Grid
Shanghai	3,365	Hua Hong, SMIC, Shanghai Jita, Fudan, ACM Research, SITRI, AMEC
Guangdong	1,993	Huawei, Tianma, TCL CSOT, Gree, CanSemi
Anhui (Hefei)	1,822	CXMT, Nexchip, Origin Quantum, USTC
Jiangsu	1,506	Visionox, JCET, China Resources Micro, Southeast Univ., Nanjing Univ.
Hubei (Wuhan)	1,473	YMTC, Changfei, XMC, HUST
Shandong	944	SiEn, Inspur, Aucma, Hisense, Shandong Univ.
Shaanxi (Xi'an)	773	Xidian Univ., Eswin
Zhejiang	629	Zhejiang Univ., Zhejiang Chuangxin
Sichuan, Liaoning, Fujian, Chongqing, Gansu	1,461	UESTC, Piotech, Sanan/Jinhua, Chongqing Xinlian, Huatian

8. Emerging technologies

Advanced packaging is the fastest-growing front in CN, ahead of every device or quantum topic. Rule: at least +40% like-for-like, and a minimum of 40 publications (CN) or 20 (TW) in the later window, on full text (codes, title and abstract) in both windows.

Office	Technology	Chinese term	Y1 → Y2	Change	Example publications (later window)
CN	Photonic quantum computing	光量子计算	15 → 66	+340%	CN122635586A photonic quantum chip packaging (Bose Quantum 玻色量子); CN122592555A thin-film lithium niobate photonic chip (Turing Quantum 图灵智算)

Office	Technology	Chinese term	Y1 → Y2	Change	Example publications (later window)
CN	3D stacking / monolithic 3D	三维堆叠	393 → 778	+98%	CN122662189A HBM device (YMTC); CN122662733A chip stacking structure (CXMT); CN122662731A DDR chip stacking (Jiangsu Huachuang)
CN	Chiplets / 2.5D / interposer	芯粒/2.5D/中介层	421 → 816	+94%	CN122662715A 2.5D micro-bump packaging (Forehope 甬矽); CN122658370A optically disaggregated HBM (Marvell); CN122641372A interposer with embedded passives (Marvell)
CN	Hybrid bonding	混合键合	404 → 735	+82%	CN122662729A wafer hybrid bonding (Chongqing Xinlian); CN122662191A memory device (YMTC); CN122662274A through-hole for hybrid bonding (Intel)
CN	Glass substrates / TGV	玻璃基板/玻璃通孔	182 → 332	+82%	CN122662697A TGV metallisation (Visionox); CN122662688A glass interconnect for embedded packages; CN122641359A through-hole glass (Jiangxi Wog)
CN	Neutral-atom quantum computing	中性原子	19 → 33	+74%	CN122656012A Rydberg optical trap (HUST); CN122549622A neutral-atom control (BAQIS)
CN	Wafer handling apparatus (ext.)	晶圆处理装置	870 → 1,310	+51%	Naura, Piotech, AMEC: domestic equipment makers
CN	Wafer bonding & layer transfer	晶圆键合与层转移	311 → 460	+48%	CN122641308A wafer-bonding chuck (Piotech); CN122662310A edge bonding (Hua Hong); CN122662581A diamond-based AlN substrate (USTB)
CN	Quantum magnetometry (ext.)	量子磁测	254 → 370	+46%	Beihang University dominates (119 in total)
CN	Compute-in-memory	存算一体	30 → 42	+40%	CN122641022A ferroelectric transistor array (USTC); CN122396216A nanoscale memristor array (Suzhou Lab)
TW	MRAM / spintronics	磁阻式記憶體	71 → 122	+72%	TW202625180A magnetic memory (Kioxia); TW202625567A (TSMC)
TW	Wafer bonding & layer transfer	晶圓接合與層轉移	76 → 130	+71%	TW202625761A substrate bonding with electrostatic chucks (Applied Materials); Soitec; Tokyo Electron

Office	Technology	Chinese term	Y1 → Y2	Change	Example publications (later window)
TW	GAA / nanosheet	環繞式閘極電晶體	40 → 63	+58%	TW202625581A GAA source/drain (Intel); TW202625585A (TSMC); TW202625579A (Tokyo Electron)
TW	3D stacking	三維堆疊	95 → 138	+45%	TW202625857A cell dies sharing a logic die (SK hynix); TW202625863A (TSMC)
TW	RRAM / memristor	電阻式記憶體	83 → 119	+43%	TW202625566A RRAM circuit (Xiamen ITRI); TW-I928723B (UMC)

Just below the +40% bar: CN fan-out packaging 454 → 625 (+38%), CN III-V 93 → 121 (+30%), TW glass substrates 31 → 41 (+32%), TW CFET 44 → 59 (+34%). Very small bases: CN topological qubits 7 → 13, TW phase-change memory 2 → 7.

New applicants (none in the earlier window, at least 15 CN / 5 TW in the later one). This is new *within the study window*, since earlier history was not queried.

- CN: Shanghai Bangxin Semiconductor 上海邦芯 (33), Turing Quantum 图灵智算 (31, photonic quantum), Chengdu Xinziguang 芯紫光 (33), CETC Institutes 58, 24, 55 and 13 (19–28 each), Beijing Ziyixin IC (24), Zhejiang Chuanghao (25), Shanghai Yaogan (20), Robert Bosch China (20), Chint New Energy (18), Harbin Institute of Technology (16), Yamaha Smart Machine (27).
- TW: Rapidus (JP, 18), Festo (DE, 10), MI Equipment (MY, 10), Nidec (8), Yamaha Robotics (8), Multiverse Computing (ES, 5), Photonic Inc. (CA, 5), Wolfspeed (US, 5), Veeco (US, 5).

The entry of several **CETC (中国电子科技集团) research institutes** points to defence-electronics institutes beginning to publish in this space.

Fastest-rising English title terms (later vs earlier window; minimum 25 CN / 10 TW mentions):

- CN: "light quantum" 5 → 29, "cold plate" 9 → 34, "packaging carrier" 7 → 27, "collaborative optimization" 7 → 26, "composite material" 34 → 91, "carrier plate" 50 → 116, "glass substrate" 44 → 99, "fan-out type" 24 → 54, "polishing solution" 49 → 99, "layout optimization" 29 → 58, "quantum measurement" 15 → 32.
- TW: "semiconductor photoresist" 5 → 23, "glass substrate" 3 → 13, "cold plate" 4 → 12, "magnetoresistive random" 5 → 13, "hybrid bonding" 13 → 22, "heat sink" 7 → 16.

Fastest-rising Google `top_terms`, CN (machine-generated key terms per document; minimum 40 in the later window): "fabricating" 15 → 124, "polishing slurry" 9 → 42, "gradient" 32 → 119, "superjunction" 19 → 58, "TGV" 27 → 81, "hardware" 28 → 81, "lattice" 20 → 50, "composite material" 38 → 93, "abrasive" 33 → 76, "semiconductor packaging" 33 →

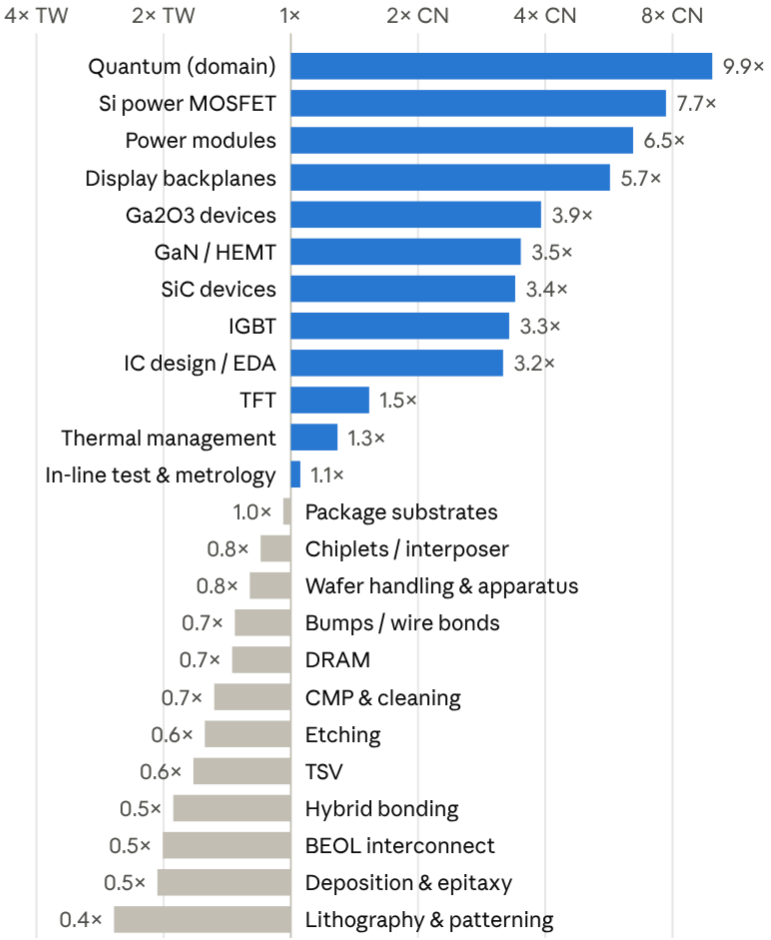
72. They confirm the title signals: glass vias (TGV), CMP consumables and superjunction power MOSFETs are rising. The TW `top_terms` mostly reflect a change in machine-translation wording ("sacrifice", "disclosure", "furthermore" from 0 to 30–50), so they are not used.

9. CN vs TW comparison

The two offices show opposite profiles. Mainland filings over-weight quantum (10×), power electronics, displays, wide-bandgap materials and EDA. Taiwan over-weights front-end process steps (lithography 2.6×, deposition 2.1×, BEOL 2.0×) and hybrid bonding/TSV, the leading-edge foundry and equipment agenda.

China leans to quantum, power and display; Taiwan to front-end process

Specialisation index = share of CN corpus ÷ share of TW corpus (log scale; 1× = same weight)



Shares use full counting (CN n = 80,062; TW n = 19,484). Blue = relatively more in CN; grey = relatively more in TW.

BigQuery, cntw_study.corpus · category shares per office

Structural indicator	CN	TW
Invention publications / families	80,062 / 79,118	19,484 / 18,824

Structural indicator	CN	TW
Top-5 applicant concentration	10.5%	30.6%
Domestic applicants	69.1%	29.6%
University or research institute involved	13.6%	2.1%
Quantum intensity (quantum domain share)	6.1% (4,852)	0.6% (118)
Advanced packaging growth (hybrid bonding + 3D + chiplets, Y1 → Y2)	1,218 → 2,329	509 → 516
Family also at EP/US/JP/KR	24.7%	68.2%
Overall volume change (like-for-like)	+2.5%	−11% (not robust)

The EP column foreseen in the prompt is left out on purpose: you asked to keep the two studies separate for now. The same core scope (PCT-excluded CN variant: 70,538 applications) allows the comparison once the EP study is run, ideally with H10P/H10W added there too.

10. Interpretation (not data)

This section reads the patterns above. Patent counts measure filing behaviour, not capability, and short windows are noisy.

Technological self-reliance (自主可控). Domestic tool makers are visible in the categories where substitution is feasible. Naura is #2 in CN wafer handling (608), and Piotech is #2 in CN CVD/ALD (319). The extension category of wafer handling apparatus grew +51%. In etching, deposition and CMP, however, Tokyo Electron and Applied Materials still lead. In lithography tools the leader is ASML (497), followed by the Chinese Academy of Sciences (358) rather than a domestic company; SMEE does not reach the top lists. This suggests substitution is advancing in handling, deposition and cleaning, while lithography remains a research-institute effort.

Export-control effects. Leading-edge logic architectures are a small share of CN filings: GAA 392, CFET 382, FinFET 155. They are carried largely by CAS, universities and foreign filers (Samsung, Intel, Qualcomm). Meanwhile mature-node and specialty fabs grow fast: Hua Hong +30%, Nexchip +78%, YMTC +29%. SMIC declines (−17%). Advanced packaging (3D stacking +98%, chiplets +94%, hybrid bonding +82%) is the strongest growth area. Huawei's profile is dominated by substrates, thermal management and encapsulation rather than transistor design. Together this is consistent with system-level integration as a route around node limits, though patent data alone cannot prove that intent.

Quantum as a state-coordinated ecosystem. Mainland quantum filings are 91% domestic and 39% academic, concentrated in the Hefei cluster (Origin Quantum, USTC, Hefei National Laboratory, China Telecom Quantum) and among state-owned operators (State Grid, China Mobile, China Telecom). QKD filings fell (−34%) while computing grew (+21%), which may indicate a shift in emphasis from secure communication to computation. Photonic (+340%) and neutral-atom (+74%) approaches are new entrants. Taiwan has almost no domestic quantum patenting (ITRI 5); its quantum filings come from foreign start-ups.

Cross-strait interdependence. Patenting runs both ways despite political tension. Taiwanese chipmakers keep a stable CN portfolio (TSMC 544 → 538). Mainland tool makers and JCET are building TW portfolios (Naura +28%). Over half of TW filings sit in families that also cover CN. 23% of US companies' TW filings name mainland-China inventors, which shows how R&D, not only markets, spans the Strait.

Implications for Europe. European applicants are a small share (3.3% of CN, 5.2% of TW). Their presence is strongest in lithography and materials (ASML, Zeiss, Merck, Soitec) and power semiconductors (Infineon, ST, Nexperia). This mirrors Europe's known strengths in equipment and power electronics. European exposure to CN quantum is minimal (161 filings), so scientific contacts in quantum may matter more than IP positioning there.

For patent statistics generally, the 2026 H01L → H10P/H10W transition can halve semiconductor counts in any study that does not update its code list. That matters for EPO-side monitoring as well.

Other readings exist. For example, the CN packaging boom may simply reflect strong demand (AI/HPC) and subsidy-driven filing behaviour rather than a strategic response to controls. CN filing volume also responds to domestic patent-quality reforms that reduced low-value filings.

11. Methodology appendix

All numbers come from the Google Patents Public Datasets on BigQuery, run in project `patent-study-513418`, dataset `cntw_study`. Every statement as executed is in the attached file `cntw_study_queries.sql`.

Queries and bytes processed (public tables)

Step	Table built	Bytes
Freshness check	—	2.08 GB

Step	Table built	Bytes
Base: CN/TW first publications in window + CPC flags	`base`	46.1 GB
IPC flags	`ipc`	9.2 GB
Titles, abstracts, claims head, keyword net	`text`	361.4 GB
Family reach (EP/US/JP/KR/CN/TW/WO)	`family_reach`	2.6 GB
Titles for H10P/H10W additions	`text_add_title`, `text_add_title2`	35.1 + 22.6 GB
CPC definitions H10P/H10W	—	0.03 GB
Second run (4 Nov, billed): Google MT text + `top_terms`	`research`	148.5 GB
Second run (4 Nov, billed): abstracts for the 26,938 additions	`text_add_abs`	218.5 GB
Rebuilds of study tables (`tx_all`, `apps`, `corpus`, `application_list`)	—	2.8 GB
Total		**≈ 849 GB**

The two queries refused for quota in the first run were run on 4 Nov under the authorised $15 budget. They billed ≈ 370 GB in total, about **$2.30** at on-demand pricing ($6.25/TB). Analysis queries on the study tables used 1–200 MB each.

Corpus definition

- **Unit:** application (first publication only; later A/B of the same application merged), plus distinct `family_id`.
- **Window:** first publication 2024-11-01 to 2026-10-31. Effective end: CN 2026-09-08, TW 2026-06-30.
- **Core scope (CPC ∪ IPC):** `^(H10D|H10B|H10N60/|H10P|H10W|H01L(21|23|25|27|29)/|G06N10/|B82Y10/|G06F30/39)`.
- **Quantum keyword net** (any text): `量子比特|量子位元|量子计算|量子計算|qubit|quantum comput`. It added 875 CN and 2 TW documents.
- **Quantum domain (final):** G06N10 or H10N60, or B82Y10 together with "quantum/qubit/量子" in the text, or the keyword net. This excluded 724 CN and 477 TW B82Y10-only documents (nanosheet/nanowire devices).

- **Extensions:** quantum comm. & sensing `H04L9/085[2-8]|H04B10/70|G01R33/03[25]|G01N24/|G04F5/14`; equipment `G03F7/|G03F1/|H01J37/32|C23C16/|H01L21/67|H10P72/`.
- **Domains:** devices/memory/processes = H10D, H10B, H01L27/29, H01L21 (non-packaging), H10P, H10W10/15. Integration = H01L23/25, H01L21/48–60, H10W20–99, G06F30/39.
- **Legacy H01L → H10 mapping:** both generations are matched in every category (e.g. etching = H01L21/3065|3213|311 or H10P50; hybrid bonding = text or H10W80; TSV = text or H10W20/2).

Technology assignment

- 65 categories. Each matches if its CPC/IPC pattern hits or its English/Chinese regex hits the title or abstract, within the domain's code scope. Process categories also require H01L21 or H10P.
- Basis recorded per document and category: CPC/IPC only · CPC/IPC + text · text within code scope.
- **Full counting:** a document counts in every matching category.
- No sampling: the whole corpus was processed. Claims are not populated for CN/TW, so claim 1 was not used.
- Trends use like-for-like windows: CN Nov 2024–Aug 2025 vs Nov 2025–Aug 2026; TW Nov 2024–Jun 2025 vs Nov 2025–Jun 2026. Since the second run, every document has title and abstract, so both periods are compared on full text.

Applicant grouping rules

- 155 ordered regex rules on "harmonised name | original-language name", first match wins. Key groupings: Samsung = Electronics + Display + SDI + SEMES; Hua Hong = Hua Hong Grace + Huahong Wuxi + Huali; Huawei = Huawei + HiSilicon; SMIC = all SMIC fabs; CAS = all CAS institutes (incl. IME, SIMIT, Suzhou Nano); JCET = JCET + STATS ChipPAC; ASE = ASE + SPIL; Kioxia = Kioxia + Toshiba Memory; Resonac incl. Showa Denko / 力森诺科; Merck KGaA incl. Versum; Visionox incl. Govisionox and Chenxian; Inspur incl. Yuanno and Yunhai Guochuang; China Telecom incl. China Telecom Quantum; State Grid incl. SmartChip; BOE incl. BOE HC SemiTek.
- Unmapped applicants keep their harmonised name; type by name tokens (大学/UNIV → university; 公司/CO LTD/INC/株式会社 → company; 研究所/研究院/实验室/INST/LAB → institute; 2–4-character personal names → individual).
- Known limits: CETC research institutes are typed "company" (name contains 公司). Fraunhofer was typed "individual" in one TW case. Harmonised names for CN are partly machine-romanised (e.g. 罗姆 → "ROEHM GMBH" is corrected to Rohm by the

map).

Origin rules

- **TW:** applicant country from `assignee_harmonized.country_code` (100% populated), cross-checked against TIPO prefixes such as 日商 (JP), 美商 (US), 大陸商 (CN).
- **CN:** applicant country is empty in the data. Order of use: (1) the parent map; (2) 株式会社 → JP; (3) INFERRED from the earliest priority office when foreign, else CN. This inference applies to 38,464 CN documents. It would mislabel a foreign firm filing first in China, or a Chinese firm claiming foreign priority.
- **Priority office** = office prefix of the earliest-dated priority claim (PCT receiving office when only a WO claim exists).
- **CN regions** are INFERRED from known headquarters of the top 50 domestic applicants only.

Deviations from the EP study

1. PCT entries kept and flagged (EP excluded them). A PCT-excluded CN variant is reported.
2. CPC ∪ IPC instead of CPC only.
3. H10P and H10W added to the core scope.
4. Quantum keyword net and the stricter B82Y10 rule.
5. Matched windows for Y1 vs Y2 instead of full years, because of the data cut-off.
6. Extension domains (quantum comm./sensing, equipment) reported separately.

Known gaps

- `top_terms` are machine-generated from machine translations. For TW they track translation wording more than technology and are not used.
- New applicants are new within the window only (no pre-2024 history).
- TW 2026 volumes fall during the classification transition, although total TW publications stay flat; not robust.
- Category regexes trade precision for recall. Spot checks found a few false positives (e.g. a consumer "light quantum" device in photonic quantum, LED "stacked" pixels in CFET).
- Extension-domain figures (quantum comm./sensing, equipment) were re-checked on full text in the second run and are unchanged.
- The full per-application list (99,546 rows, all CSV columns requested, plus `top_terms`) is in BigQuery as `patent-study-513418.cntw_study.application_list` (≈150 MB). It cannot be pulled through this session: the BigQuery connection returns rows into the

conversation, and a file export needs a Cloud Storage login the session does not have. Export it from the BigQuery console: open the table → Export → Export to GCS (CSV), or query it and use Save results → Google Drive.

As-of: data queried 3 Nov 2026; abstracts, machine translations and top_terms added and all figures recomputed on 4 Nov 2026.